

AI accountability, in an ocean of responsibility-gaps and opacity

Neil Lawrence once said: “We have built computers that make errors, but we have not built machines that understand or suffer the consequences of those errors” (Lawrence). This insight captures a central dilemma of artificial intelligence: while AI systems increasingly shape human lives, they remain incapable of bearing moral responsibility for the harms they may cause.

A world-changing revolution is happening as a result of Artificial Intelligence (AI) diversification. Nowadays, it is integrated as part of our daily lives, in the form of LLMs as chatbots like ChatGPT or Gemini, to assist in tasks like e-mail writing or code generation. This innovative tool has also reached into the decision-making process of sectors like healthcare, economics, and even law. Although AI holds a humongous amount of computational power for prediction and decision-making processes, an uncertainty arises: If an AI makes a mistake, who should be responsible? This question unfolds many opinions related to how AI accountability in cases of harm originated by the algorithms behind AI should be addressed. Furthermore, as a result of this issue, ethical concerns about the opaque nature of these algorithms, known as the “black box problem,” appear (Mittelstadt et al.).

In order to find a resolution, it's crucial to understand the core problem: traditional responsibility frameworks- product liability, corporate responsibility, negligence, and strict liability- were not designed for autonomous, opaque, and learning systems (Kumar). An incident that illustrated this dilemma happened in 2019 when the self-driving software of a Tesla car caused the death of a pedestrian. In this instance, the legal system encountered a "Responsibility Gap" where fault was fractured across three distinct entities: a distracted driver, a software system that failed to brake, and a corporation that marketed the technology as "Autopilot" despite its operational limits. Because traditional negligence requires a single "proximate cause," the resulting legal friction demonstrates a desperate attempt to retroactively fit distributed algorithmic failure into a legal box designed for human-only error. This highlights the framework's failure: it cannot easily distinguish whether the "negligence" lies in the driver's lack of focus, the engineer's lack of geofencing, or the CEO's choice of marketing.

This essay proposes a multi-stakeholder, distributed responsibility model, designed to assign accountability asymmetrically across stakeholders according to control, foreseeability, benefit, and severity of harm—placing primary design responsibility on corporations, structural oversight on regulators, and context-dependent duties on professional users. Due to AI's lack of moral agency and with the aim to bridge the “Responsibility Gap,” adapting existing legal principles and introducing new ethical considerations is essential (Noh).

The Nature of AI Mistakes and the "Responsibility Gap"

Understanding the spectrum of harms caused by AI is essential for any principled accountability framework. Errors range from low-impact inconveniences to severe economic,

reputational, and life-threatening failures (Porter et al.). A defining challenge is the "black box" nature of deep learning models, whose internal reasoning processes are opaque even to their designers (Mittelstadt et al.). When an AI system produces a harmful outcome, neither affected individuals nor courts can easily determine why the decision occurred, making responsibility difficult to assign.

The consequences of this opacity are illustrated by Joy Buolamwini's *Gender Shades* project, which revealed that commercial facial recognition systems performed with near-perfect accuracy on lighter-skinned men while exhibiting error rates approaching 35% for darker-skinned women (Buolamwini and Gebru). Under traditional negligence law, victims must demonstrate a breach of duty or intent—standards ill-suited to harms generated by statistical correlations embedded within opaque systems. As a result, victims of algorithmic discrimination are often left without a clear path to redress, exposing a structural responsibility gap (Kasar).

AI cannot hold accountability in harm scenarios because moral agency requires attributes fundamentally linked with human nature, such as autonomy and free will (Rebera). Modern systems lack the internal freedom vital for sensible moral responsibility; without a legitimate understanding of moral principles like justice or honesty, AI remains limited to computational processes rather than independent moral reasoning. To explain the limitation of traditional frameworks, the concept of "shared responsabilization" offers a more adaptive approach (Lang et al.). This frames the gap not as an unfillable void, but as a challenge in attribution that can be met through collective accountability.

Shortcomings of Existing Frameworks in the AI Context

Traditional legal frameworks struggle to accommodate AI systems due to their opacity and adaptability. Product liability is adequate for physical products, but identifying a "defect" in a continuously evolving AI system is problematic (Tatipamula). The Boeing 737 MAX crashes demonstrated that software defects can constitute design flaws, but the case also revealed how complexity enables manufacturers to deflect blame toward operators.

Corporate responsibility encourages ethical guidelines, yet attributing specific errors to a single entity remains challenging. Amazon's abandoned AI recruiting tool illustrates this: the system penalized applications associated with women because it was trained on historically male-dominated resumes (Chang). The harm resulted from systemic failures in data selection rather than a single actor. Similarly, negligence requires proving a "breach of duty," which the "black box" nature of AI impedes. IBM Watson for Oncology produced unsafe treatment recommendations because it was trained on synthetic cases rather than real patient data (Ross). The opacity of Watson's reasoning made it nearly impossible to trace the harm to a specific human decision, allowing technical complexity to obscure what would otherwise constitute a breach of care (Choi and Lam).

Strict liability for high-risk AI is often seen as the most promising framework because it removes the requirement to prove fault (Mao). However, determining a single "manufacturer" can be ambiguous due to the complex value chain involving data providers, developers, and testers.

Towards a Distributed and Non-Anthropocentric Responsibility Model

A model of “shared responsabilization” involves allocating accountability among three tiers: firstly, manufacturers and developers for ethical design and Explainable AI (XAI); secondly, users and operators for oversight; and thirdly, regulators for establishing guidelines (Lang et al.; Noh).

Manufacturers are primarily accountable for knitting safety measures from the initial stages. This requires the integration of XAI to penetrate the 'black box' and ensure processes are auditable (Mittelstadt et al.). Users—particularly in healthcare and law—retain a duty of human oversight; blind reliance on AI is ethically indefensible, as seen in the IBM Watson case (Choi and Lam). Regulators must transition toward frameworks like the EU AI Act that utilize System Safety Theory to distribute fault across the entire value chain (European Union).

Critics argue that a shared model leads to a diffusion of culpability. However, this concern conflates distributed fault with legal ambiguity. To prevent a bypass of responsibility, the model must incorporate a "reverse burden of proof," requiring developers to demonstrate safety measures rather than forcing the victim to untangle proprietary code (Fraser and Suzor).

Conclusion and Future Outlook

Is there an ultimate solution to know who should be responsible if AI makes a mistake? No single framework will address all scenarios of harm, but establishing distributed responsibility is the most viable path toward justice. Adaptive governance, legal innovation, and ethical reflection are the vital components to build a society that uses AI in a responsible way.

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